

Talk Like a Structured Graph? Structural Statistics Help Small LLMs Mainly by Stating the Answer

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Abstract

Graph models gain from structural encodings such as node degree and random-walk return probabilities. We ask whether a frozen language model reading a graph as text gains when the same statistics are written as sentences in a *primer* before GraphQA’s incident encoding. In paired tests at several edge densities, for two model sizes with thinking off and on, we compare each primer with no primer, a structure-free filler and the model’s thinking mode. Mostly, statistics do not improve how the model computes; they help by stating the answer. Stated degrees raise node-degree accuracy by 7.5 to 12.2 points where a model counts unreliably, and cost the one model that already counts correctly 6.5 to 8.0. Other statistics have mostly small, model-dependent effects and often change the same answers as the filler. They narrow the gap to thinking without closing it, except one sentence that makes the model restate the queried node’s line, matching thinking on one task. With thinking on, models check stated values against their own count, and much of a primer’s effect is on finishing within the budget. Accuracy can also mislead: without thinking, stated degrees can lower cycle-detection accuracy yet raise the share of validly argued answers.

1 Introduction

Structural encodings, such as node degree and random-walk return probabilities, are a standard ingredient of graph neural networks and graph Transformers (Dwivedi et al., 2022; Rampásek et al., 2022). A language model that reads a graph as text must instead recover such quantities itself, and its accuracy depends on how the graph is written (Fatemi et al., 2024; Wang et al., 2023). Statistics are cheap to precompute, so an obvious aid is to state them in the prompt. We ask: *when a frozen language model reads a textual graph, do explicit structural statistics improve execution of graph queries?* By *execution* we mean computing

the answer from the graph’s edges, not copying it from elsewhere in the prompt.

A primer can raise accuracy by stating the answer, by adding information that narrows it, or by changing how the model reads the graph; only the last is better execution. In an Erdős-Rényi graph every edge is independent of the others, so beyond a node’s degree, clustering and return probabilities carry almost no information about GraphQA’s answers (§4.1). A gain from them therefore reflects how the model reads the prompt, not new information. We measure each primer’s effect in both modes against three references: no primer; a structure-free *filler*, for what added text does on its own; and, for the plain mode, the same model thinking without a primer. The last asks whether a preamble can stand in for reasoning that costs 6.2-17.1 times the output tokens on node degree. The degree primer, which states the answer to node-degree and edge-count queries, is a positive control.

Specifically: does a primer that states the answer help, and where? Do statistics that do not state it help beyond what any added text does? Can a primer bring a model’s plain mode level with its thinking mode? And do the responses show the primer being used? In short, explicit statistics mostly do not improve execution.

Contributions. (i) A stated answer substitutes for execution: it helps where the model’s own count is unreliable and hurts where it is reliable (§4.2). (ii) Measured against a graph-blind solver and a structure-free filler, other statistics have mostly small effects whose sign depends on the model (§4.3). (iii) One sentence that changes the procedure matches thinking, on one task (§4.4). (iv) Responses show stated values being used but not other statistics, and correct answers can rest on cycles the graph lacks (§4.6).

2 Related Work

Graphs as text. GraphQA (Fatemi et al., 2024) and NLGraph (Wang et al., 2023) show that LLM accuracy on graph queries depends on encoding, task and graph structure, and NLGraph’s build-a-graph and algorithmic prompts change the procedure by instruction. We keep GraphQA’s tasks and encoding and change only a preamble.

Structure for LLMs. Structure can be passed as learned soft tokens (Perozzi et al., 2024) or as motif descriptions, which help on some tasks (Das et al., 2024). Structural descriptions in a prompt, however, need not be read as topology (Huang et al., 2024). We state the encodings of graph learning as sentences to a frozen model and control for what they state.

Controls and faithfulness. Added or irrelevant text changes LLM answers (Shi et al., 2023; Levy et al., 2024), as does formatting (Sclar et al., 2024). Partial-input baselines expose answers a dataset gives away (Feng et al., 2019), as our graph-blind solver does for primers. A right answer can rest on a wrong route (Geirhos et al., 2020), and a model’s stated reasoning need not reflect the computation behind it (Turpin et al., 2023), so we treat responses as claims to check.

3 Method

Models and graphs. Qwen3-1.7B and Qwen3-4B (Qwen Team, 2025), each with thinking off (plain, P) and on (T), make four *arms*; decoding is zero-shot and greedy, one response per prompt (budgets in Appendix A). A pilot on the published GraphQA graphs with Gemma 4 (Google DeepMind, 2026) and Qwen3-8B/14B left most arms near ceiling (Appendix C), and even at 40 nodes Qwen3-8B answers 96.5-99.8% of node-degree queries correctly. We therefore use harder graphs and smaller models, so that accuracy without a primer leaves room to measure a change in either direction. Our graphs are Erdős-Rényi $G(n, p)$ (Erdős and Rényi, 1959) with $n = 40$ at four edge densities up to $p = .50$ (the *main sweep*) and, for node degree and edge existence, three more up to $p = .85$. Every condition and arm sees the same 100 graphs per task and density, so every comparison is paired.

Tasks. GraphQA’s six tasks in its incident encoding (Figure 1). Four have answers that vary:

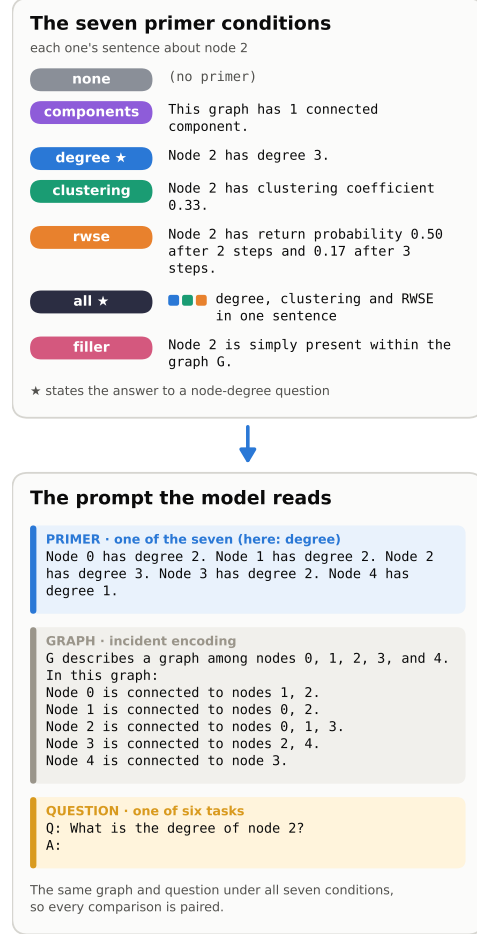


Figure 1: How a primer is used. Top: the seven conditions, each with its sentence about node 2 of a five-node example (the study’s graphs have 40 nodes); ★ marks the two that state a node-degree answer. Bottom: the prompt the model reads, one primer followed by GraphQA’s incident encoding and the question.

node degree, connected nodes (the queried node’s neighbors, exact set match), edge count and edge existence. Two cannot measure reading at 40 nodes: the encoding’s first line lists all 40 ids and every graph has a cycle, so node count is always 40 and cycle check always yes (Appendix B). Cycle check serves only to ask what correct answers rest on.

Primers and controls. Before the encoding we place no primer, one sentence per node giving its degree, local clustering coefficient or return probabilities after 2 and 3 steps (RWSE; Dwivedi et al., 2022), or *all* three (Appendix A). Two diagnostics follow the same format: a sentence giving the number of connected components, and the filler, which names every node and states no structure. The filler is about as long as the clustering primer, longer than degree and shorter than RWSE and all

Task	Arm	none	degree	clustering	RWSE	all	components	filler
Degree	1.7P	60.50	+7.5 ^f	+3.0	+4.0	+10.2 ^f	+1.8	-0.5
	1.7T	76.25	+8.8 ^f	+1.5 ^f	+2.5 ^f	+12.2 ^f	-1.0	-3.0
	4P	99.25	-6.5 ^f	-0.5	-2.2	-8.0 ^f	+0.2	-0.5
	4T	97.00	+2.5	+1.8	-1.2 ^f	+0.5	+0.2	+2.0
Neighbors	1.7P	72.50	+3.5 ^f	+1.2 ^f	+1.0 ^f	+2.0 ^f	+0.8 ^f	-6.2
	1.7T	78.75	+6.2	+2.0	+0.2	+6.2	+2.5	+2.5
	4P	86.00	-3.0 ^f	+1.0 ^f	+0.0 ^f	-2.8 ^f	+9.2 ^f	-12.5
	4T	95.50	+0.5	-0.8	+0.5	-1.2	+0.5	+1.2
Edge count	1.7P	1.25	+3.0 [†]	-0.5 [†]	-0.8 [†]	+0.8 [†]	-1.0 [†]	-0.5 [†]
	1.7T	15.75	-3.2 [†]	-1.2 [†]	-3.2 [†]	+5.0 [†]	-0.8 [†]	-1.5 [†]
	4P	2.00	+27.2 ^f	-1.2 ^f	-1.5 ^f	+3.8	-1.2 ^f	+1.2
	4T	20.00	+8.5 [†]	+24.8 [†]	+10.2 [†]	+1.8 [†]	+24.8 [†]	+12.2 [†]
Edge exists	1.7P	70.50	+4.0 ^f	+3.8 ^f	+5.0 ^f	+11.5 ^f	-8.2 ^f	-15.8
	1.7T	99.00	-2.5	-1.0	-1.2	-1.8	-0.8	-0.5
	4P	90.50	-4.8	+4.0 ^f	-10.0	-10.0	-2.8 ^f	-7.0
	4T	99.75	+0.0	-0.8	-1.2	-1.0	-0.5	+0.0

Table 1: Primary result: correct share without a primer (% of all prompts) and the paired change under each primer (points), on the same 400 graphs per row ($p \leq .50$). Shaded: the primer states the answer. Bold: $q < .05$ against no primer (BH within arm, task and control); ^f: $q < .05$ against filler; [†]: at least 15% truncation on either side (unmarked). Intervals and the constant tasks: Table 5; truncation: Table 8; balanced accuracy: Table 6. P/T: plain/thinking.

three (Appendix Table 4). The graph-blind solver answers from the primer text alone.

Outcomes and statistics. Each response is *correct*, *wrong* or *truncated* (it used its whole budget); shares use all responses, and an effect is the paired change in the correct share, in points. Cells with at least 15% truncation on either side are flagged ([†]) and left out of pooled summaries. Edge existence is also scored by balanced accuracy, because the share of queried pairs that are edges rises from 10% to 85% with density. The primary analysis is Table 1: each primer against no primer and against filler on the same 400 main-sweep graphs. It uses exact McNemar tests (McNemar, 1947), bootstrap intervals over graphs and Benjamini-Hochberg correction (Benjamini and Hochberg, 1995) within each (arm, task, control) family; $q < .05$ is significant. Plain arms against thinking (Table 2) use the same tests on the same graphs, and we call them equivalent when the 90% interval of their difference lies within ± 3 points, about the 4B arms’ median detectable gain. Analyses of mechanism are secondary and name their own test. Two limits of these tests matter for reading the results. *Power*: over 400 pairs the median detectable gain is 3.2 to 5.7 points, so a non-significant effect is unresolved, not zero (Appendix A). *Noise*: generating the same prompts twice changes the outcome of 5.3%, so we compare net paired changes, never counts of changed answers.

Response readers. Rule-based readers record whether a response reads a stated value off (a *look-*

up) or lists the neighbors, whether it reports a conflict with the primer, and which cycles and edges it asserts, checked against the graph. The conflict readers agree with blind hand labels on 90-100% of 226 items; the others were spot-checked by an LLM under human supervision (Appendix A).

Follow-ups. Qwen3-1.7B node degree, plain and thinking, 400 graphs per density at all seven densities, a fresh seed, and a 20-160-node grid at mean degree 8 and 16 (also Qwen3-8B).

4 Results

4.1 What each primer states

The graph-blind solver answers node degree and edge count perfectly from the degree primer or all three statistics, by reading the queried degree or halving the sum of the stated degrees. Every other primer barely informs it: at most 21.0% on node degree, 10.4% on edge count and 74.6% on edge existence, against 14.4%, 3.0% and 73.5% with no primer (Appendix Table 3). The shaded cells of Table 1 are these answer-stating primers; we call every other cell a *side* effect.

4.2 Stated answers are read off

The degree primer raises node-degree accuracy for plain and thinking Qwen3-1.7B by 7.5 and 8.8 points and lowers it by 6.5 for plain Qwen3-4B, which already counts correctly. The gain is largest where a model is right about half the time and becomes a loss where it is right most of the time (Figure 2). The same pattern holds within one

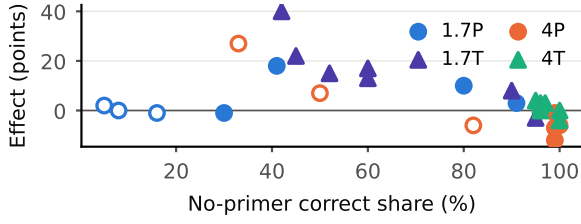


Figure 2: The degree primer’s effect on node degree against the no-primer correct share, per arm and density. Hollow: plain arm at $p \geq .65$ (smaller budget).

model. The degree primer costs plain Qwen3-4B 11.8 points on the queried nodes with the lowest degrees in their graph, and gains 10.6 on those with the highest, which have the most neighbors to count.

Reading a value off is itself fallible. Across all seven densities, where the answer is stated, plain Qwen3-1.7B gets it right only 43% of the time. Plain Qwen3-4B stops listing the node’s neighbors and reads the value off in 62% of responses at $p = .50$. A third of its wrong look-ups copy the degree printed one sentence away, more often than chance predicts (Appendix B).

4.3 Side statistics have mostly small, model-dependent effects

Filler alone changes the outcome of 6.2-18.6% of questions per arm, largely the same questions the primers change. Part of this is generation noise, though questions that flip on an identical rerun carry only a minority of each condition’s changes (Appendix B).

Significant side effects differ in sign across statistics and models. On plain Qwen3-4B’s edge existence, clustering raises balanced accuracy and RWSE lowers it ($q \leq .018$). All three statistics raise plain Qwen3-1.7B’s by 7.8 points but lower plain Qwen3-4B’s by 6.5 (Appendix Table 6). The largest side gain without the truncation flag, +9.2 on plain Qwen3-4B’s neighbor sets, comes from the component sentence, which is the same for every connected graph. With it the model restates the queried node’s line in nearly every response (98.5%, against 63.0%): a change of procedure without information. Where filler costs accuracy, clustering and RWSE avoid most of that cost, scoring 12.5 to 20.8 points above it. At high density, where they barely vary, their gains are best read as presentation effects (Appendix B).

The clearest mechanism is on edge existence. Plain Qwen3-1.7B’s false alarms climb with shared

Task	Model	Plain	Best primer	Think.	Left	CI/RW.
Degree	1.7B	60.50	all 70.75	76.25	+5.5	+11.8
	4B	99.25	components 99.50	97.00	-2.5	-1.8
Neighbors	1.7B	72.50	degree 76.00	78.75	+2.8	+5.0
	4B	86.00	components 95.25	95.50	+0.2	+8.5
Edge count	1.7B	1.25	degree 4.25	15.75 [†]	+11.5	+15.0
	4B	2.00	degree 29.25	20.00 [†]	-9.2	+19.2
Edge exists	1.7B	70.50	all 82.00	99.00	+17.0	+23.5
	4B	90.50	clustering 94.50	99.75	+5.2	+5.2

Table 2: Plain mode against the same model thinking without a primer (% correct, $p \leq .50$). *Left*: thinking minus the plain arm’s best score; *CI/RW.*: the same for the better of clustering and RWSE. Bold: $q < .05$. Chosen on held-out graphs, the best primer is the same in every row. [†]: thinking truncates on most prompts (Table 8).

neighbors, from 1.7% of non-edges that share none to 65.9% of those sharing four or more. In $G(n, p)$ shared neighbors say nothing about whether a pair is joined, so the models appear to read two ids on the same lines as adjacency. A statistic changes the stated reason more than the decision: answers claiming a non-edge is an edge fall from 13.7% to at most 4.4%, but false alarms fall only from 40.3% to 35.2% under clustering.

4.4 Primers against the thinking mode

Where thinking leads (Table 2), one primer brings the plain mode level with it: the component sentence on Qwen3-4B’s neighbor sets, equivalent to thinking within ± 3 points. Elsewhere primers narrow the gap without closing it. All three statistics cut plain Qwen3-1.7B’s node-degree gap from 15.8 points to 5.5. Under the degree primer its neighbor sets are not significantly below thinking, but not provably level either. Clustering and RWSE close less, and leave the plain mode significantly below thinking wherever thinking leads. The degree primer lifts plain Qwen3-4B’s edge count above thinking. Thinking there is right on 97.6% of the answers it finishes, though, so this compares budgets, not ability, and the stated degrees already give the edge count.

4.5 Primers in the thinking mode

Near ceiling, thinking Qwen3-4B moves by at most 2.5 points under any primer on node degree, edge existence and neighbor sets. Thinking Qwen3-1.7B gains from stated degrees (Table 1) but checks them against its own count instead of reading them off. It keeps listing neighbors, and when its count conflicts with the stated value, it

ends on the stated value in 66% of cases. Checking costs budget: across all seven densities, its node-degree responses that run out of tokens rise from 1.1% to 7.6% under the degree primer. Where the budget binds, much of any primer’s effect is on finishing. Most of thinking Qwen3-4B’s changed answers involve a response that ran out of tokens, and on its edge count even filler gains by lowering truncation (Table 8).

4.6 What responses rest on

Stated values are visibly used (§4.2, §4.5); here we ask whether a primer improves reading the graph itself. A model that reads the queried node’s line gets both its degree and its neighbor set right. Clustering and RWSE never move this joint share significantly; the degree primer cuts plain Qwen3-4B’s from 85.75% to 76.50%, while the component sentence lifts it to 95.00% (Appendix B).

Cycle check needs no computation at 40 nodes, but a response can justify its “yes” validly: by a real cycle, or by the edge count. Without a primer, a valid argument backs only 34.2% and 58.5% of the plain models’ prompts, against 78.0% and 95.8% with thinking. Plain Qwen3-1.7B rests 43.5% of its correct answers on an invented cycle, one with a step that is not an edge. The degree primer raises its valid share from 34.2% to 50.2% yet lowers its accuracy by 17.5 points. The plain models switch to arguing from the number of edges (Table 7), a valid route that, like a look-up, need not read the edges. RWSE, all three statistics and filler lower plain Qwen3-4B’s valid share.

The one side effect that replicates, clustering on plain Qwen3-1.7B’s node degree, is small, specific to that model and to low density, and shows little sign of using the stated values (Appendix C).

5 Discussion

As in prior work, stated structure need not be read as topology (Huang et al., 2024), and added text alone moves answers (Shi et al., 2023; Levy et al., 2024); learned structural tokens (Perozzi et al., 2024) may reach what sentences do not.

Two lessons follow from these models. First, whether to add a primer depends on how reliably the model computes the quantity unaided: in our data stated values helped only where that computation was unreliable. Second, accuracy misstates execution in both directions. Raw edge-existence accuracy rewards saying yes on dense graphs: from

$p = .50$ to $p = .85$, plain Qwen3-1.7B’s rises from 54% to 86% while its balanced accuracy stays near chance (53%). And the degree primer lowers cycle-check accuracy while raising the share of valid answers.

6 Future Work

Four tests follow. First, primers that state wrong values, or omit the queried node’s own value, would show whether a model reads a statistic off, checks it or ignores it. Second, randomized node labels and preamble order, against token-matched neutral text, would separate presentation effects, such as the component sentence’s, from information. Third, graphs where the statistics carry information (triadic closure makes clustering predict edges; acyclic graphs give cycle check two answers), on a second model family with a larger thinking budget, would show how far the findings reach. Fourth, a primer’s benefit could be predicted from unaided accuracy (§5) and weighed against giving the model a graph tool or code.

7 Conclusion

Explicit structural statistics mostly do not improve execution for two small Qwen3 models: they help mainly by stating the answer, which the plain models read off rather than compute. Since accuracy can misstate execution, evaluations of LLMs on graphs should check what an answer rests on, not only whether it is right.

Limitations

Scope: two small models from one family (and Qwen3-8B in one follow-up); Erdős-Rényi graphs with 40 nodes (20-160 in one follow-up); one encoding, wording and position per primer; zero-shot; one token budget. *Controls:* the filler matches only the clustering primer’s length, in characters. No primer carries shuffled or wrong values, and no instruction-only baseline tests procedural effects. Two tasks have a constant answer, and node ids 0-9 are both listed first and single-digit, so a node’s position and its id length are confounded. *Measurement:* generation noise (§3), and readers other than the conflict readers were checked by an LLM, not hand-labelled. Under one BH over all 264 tests, every bold cell of Table 1 but one stays significant ($q = .057$), and the clustering effect was found by a screen but replicates on new graphs at low density (Table 9).

References

- Yoav Benjamini and Yoel Hochberg. 1995. [Controlling the false discovery rate: A practical and powerful approach to multiple testing](#). *Journal of the Royal Statistical Society: Series B (Methodological)*, 57(1):289–300.
- Debarati Das, Ishaan Gupta, Jaideep Srivastava, and Dongyeop Kang. 2024. [Which modality should I use - text, motif, or image? : Understanding graphs with large language models](#). In *Findings of the Association for Computational Linguistics: NAACL 2024*, pages 503–519.
- Vijay Prakash Dwivedi, Anh Tuan Luu, Thomas Laurent, Yoshua Bengio, and Xavier Bresson. 2022. [Graph neural networks with learnable structural and positional representations](#). In *International Conference on Learning Representations*.
- Paul Erdős and Alfréd Rényi. 1959. [On random graphs I](#). *Publicationes Mathematicae Debrecen*, 6(3-4):290–297.
- Bahare Fatemi, Jonathan Halcrow, and Bryan Perozzi. 2024. [Talk like a graph: Encoding graphs for large language models](#). In *International Conference on Learning Representations*.
- Shi Feng, Eric Wallace, and Jordan Boyd-Graber. 2019. [Misleading failures of partial-input baselines](#). In *Proceedings of the 57th Annual Meeting of the Association for Computational Linguistics*, pages 5533–5538.
- Robert Geirhos, Jörn-Henrik Jacobsen, Claudio Michaelis, Richard Zemel, Wieland Brendel, Matthias Bethge, and Felix A. Wichmann. 2020. [Shortcut learning in deep neural networks](#). *Nature Machine Intelligence*, 2(11):665–673.
- Google DeepMind. 2026. Gemma 4 model cards (gemma-4-E4B-it, gemma-4-12B-it). <https://huggingface.co/google>.
- GraphTalk Contributors. 2026. GraphTalk: Structural primer experiments and results. <https://github.com/GalBrk/GraphTalk/tree/results-sot>. Results-sot branch, commit 3cf63ecb85382d0cef2ef7082bflccb6e435f730.
- Jin Huang, Xingjian Zhang, Qiaozhu Mei, and Jiaqi Ma. 2024. [Can LLMs effectively leverage graph structural information through prompts, and why?](#) *Transactions on Machine Learning Research*.
- Mosh Levy, Alon Jacoby, and Yoav Goldberg. 2024. [Same task, more tokens: the impact of input length on the reasoning performance of large language models](#). In *Proceedings of the 62nd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 15339–15353.
- Quinn McNemar. 1947. [Note on the sampling error of the difference between correlated proportions or percentages](#). *Psychometrika*, 12(2):153–157.
- Bryan Perozzi, Bahare Fatemi, Dustin Zelle, Anton Tsitsulin, Mehran Kazemi, Rami Al-Rfou, and Jonathan Halcrow. 2024. [Let your graph do the talking: Encoding structured data for LLMs](#). *arXiv preprint arXiv:2402.05862*.
- Qwen Team. 2025. [Qwen3 technical report](#). *arXiv preprint arXiv:2505.09388*.
- Ladislav Rampášek, Michael Galkin, Vijay Prakash Dwivedi, Anh Tuan Luu, Guy Wolf, and Dominique Beaini. 2022. [Recipe for a general, powerful, scalable graph transformer](#). In *Advances in Neural Information Processing Systems*, volume 35, pages 14501–14515.
- Melanie Sclar, Yejin Choi, Yulia Tsvetkov, and Alane Suhr. 2024. [Quantifying language models’ sensitivity to spurious features in prompt design or: How I learned to start worrying about prompt formatting](#). In *The Twelfth International Conference on Learning Representations*.
- Freda Shi, Xinyun Chen, Kanishka Misra, Nathan Scales, David Dohan, Ed H. Chi, Nathanael Schärli, and Denny Zhou. 2023. [Large language models can be easily distracted by irrelevant context](#). In *Proceedings of the 40th International Conference on Machine Learning*, volume 202 of *Proceedings of Machine Learning Research*, pages 31210–31227.
- Miles Turpin, Julian Michael, Ethan Perez, and Samuel Bowman. 2023. [Language models don’t always say what they think: Unfaithful explanations in chain-of-thought prompting](#). In *Advances in Neural Information Processing Systems*, volume 36, pages 74952–74965.
- Heng Wang, Shangbin Feng, Tianxing He, Zhaoxuan Tan, Xiaochuang Han, and Yulia Tsvetkov. 2023. [Can language models solve graph problems in natural language?](#) In *Advances in Neural Information Processing Systems*, volume 36, pages 30840–30861.

A Method Details

Statistics. With adjacency matrix A and transition matrix $P = D^{-1}A$, the per-node statistics are $d_v = \sum_u A_{vu}$, $c_v = 2t_v/(d_v(d_v - 1))$ and $r_v^{(k)} = (P^k)_{vv}$ for $k \in \{2, 3\}$, where t_v counts edges among v ’s neighbors and $c_v = 0$ when $d_v < 2$. Clustering and return probabilities are printed to two decimals, one sentence per node, in node order (Table 4).

Why these features. Degree and RWSE are the structural encodings graph Transformers use (Rampásek et al., 2022). Local clustering adds how a node’s neighbors connect. The component count is a single global fact that, with the node and edge counts, decides whether a cycle exists. Laplacian encodings are left out because their sign and basis are arbitrary and cannot be stated as facts (Dwivedi et al., 2022).

Budgets and power. Each response may use up to 8,192 new tokens, and 2,048 for the plain arms at $p \geq .65$, where no arm truncates more than 2.5% of its responses. The main sweep and its extension hold 84,000 responses on 700 graphs. In one 100-graph cell the median smallest detectable paired effect is 7.4 points, and 13.7 where side information meets intermediate accuracy.

Graph-blind solver. It reads only the primer text, with 16 exact rules, 1 heuristic and 8 rules fitted on graphs disjoint from those scored (Table 3).

Primer	Degree	Neighbors	Edge count	Edge exists
none	14.4	0.0	3.0	73.5
degree	100.0	0.1	100.0	74.6
clustering	14.4	0.0	10.4	72.7
RWSE	21.0	0.1	3.0	55.4
all	100.0	0.8	100.0	74.1
components	14.4	0.0	3.0	73.5
filler	14.4	0.0	3.0	73.5

Table 3: Graph-blind solver accuracy (%) from the primer text alone, 40-node graphs, on the four tasks whose answer varies.

Response readers. A response is a look-up when it answers without restating the queried node’s neighbor list, and lists the neighbors when it gives five or more one per line. The cycle and edge readers check every asserted step against the prompt’s graph.

Scoring and provenance. The answer extractor parses 99.84% of responses. Set-F1 on neighbor sets is secondary because it gives partial credit: it

Primer	Mean chars	Information
degree	891	Degree for each node
clustering	1,629	Local clustering for each node
RWSE	2,949	Two rounded return values per node
all	4,691	Degree, clustering and RWSE
components	37	Number of connected components
filler	1,829	Node names, no graph statistic

Table 4: Mean preamble length in characters over the main-sweep graphs.

moves in the same direction as exact match in 3 of 5 large contrasts, and by 0.11 to 0.17 as much. Balanced-accuracy changes use a paired permutation test with Benjamini-Hochberg adjustment over the six conditions within an arm. Tagged results come from GraphTalk’s `results-sot` branch (GraphTalk Contributors, 2026): each tag names a block of a script output cited by the results documents in `docs/`.

B Detailed Results

Tables 5 and 8 complete Table 1 with 95% intervals, truncated shares and the two constant-answer tasks. The paragraphs below give the numbers Section 4 summarizes, in its order.

Stated answers (§4.2). The gain follows the room the model leaves (Figure 2). In node-degree cells answered 25-50% and 50-75% correctly without them, answer-stating primers gain 16.1 and 12.9 points; at 75-90% they lose 5.8. These bands hold on held-out halves of the graphs and with the truncation flag at 10% or 50% (at least +12.5 in every case). An effect mostly follows how well the model does unaided. At a fixed accuracy without a primer, density adds little ($p = .28$ for answer-stating primers, $p = .85$ for side statistics), except that side statistics’ effect on node degree falls with density ($p = .006$). One model falls on both sides. The degree primer costs plain Qwen3-4B 12.0 points at $p = .50$, where it answers 99.0% correctly unaided, and gains 27.0 at $p = .85$, where it answers 33.0% (both $q < .01$). Stated answers are not always taken: across all seven densities, where the solver scores 100%, node-degree accuracy under the degree primer is 43% and 81% for plain Qwen3-1.7B and Qwen3-4B, and 79% and 98% with thinking. At high density, where plain Qwen3-1.7B is weakest, the stated degree barely

Task	Arm	degree	clustering	RWSE	all	components	filler
Degree	1.7P	+7.5 [+3.0, +11.8]	+3.0 [-1.5, +7.5]	+4.0 [-0.5, +8.5]	+10.2 [+5.2, +15.2]	+1.8 [-2.2, +5.8]	-0.5 [-4.2, +3.5]
	1.7T	+8.8 [+4.2, +13.2]	+1.5 [-2.5, +5.5]	+2.5 [-2.0, +6.8]	+12.2 [+8.0, +16.5]	-1.0 [-4.2, +2.5]	-3.0 [-7.0, +0.8]
	4P	-6.5 [-9.2, -4.0]	-0.5 [-2.0, +0.8]	-2.2 [-4.2, -0.5]	-8.0 [-11.0, -5.2]	+0.2 [+0.0, +0.8]	-0.5 [-2.0, +1.0]
	4T	+2.5 [+0.8, +4.2]	+1.8 [-0.2, +3.8]	-1.2 [-3.8, +1.0]	+0.5 [-1.5, +2.8]	+0.2 [-1.8, +2.5]	+2.0 [+0.5, +3.8]
Neighbors	1.7P	+3.5 [-0.5, +7.2]	+1.2 [-3.0, +5.5]	+1.0 [-3.2, +5.2]	+2.0 [-2.5, +6.5]	+0.8 [-2.5, +4.5]	-6.2 [-10.5, -2.2]
	1.7T	+6.2 [+2.8, +10.0]	+2.0 [-2.0, +6.0]	+0.2 [-3.5, +4.0]	+6.2 [+2.5, +10.0]	+2.5 [-0.8, +6.0]	+2.5 [-1.0, +6.0]
	4P	-3.0 [-7.0, +1.0]	+1.0 [-2.8, +4.8]	+0.0 [-3.8, +4.0]	-2.8 [-7.0, +1.8]	+9.2 [+6.0, +12.8]	-12.5 [-17.2, -7.8]
	4T	+0.5 [-2.2, +3.2]	-0.8 [-3.2, +1.8]	+0.5 [-1.8, +3.0]	-1.2 [-3.8, +1.2]	+0.5 [-1.8, +2.8]	+1.2 [-1.0, +3.5]
Edge count	1.7P	+3.0 [†] [+0.8, +5.2]	-0.5 [†] [-2.0, +1.0]	-0.8 [†] [-2.0, +0.5]	+0.8 [†] [-1.0, +2.5]	-1.0 [†] [-2.2, +0.0]	-0.5 [†] [-1.8, +0.8]
	1.7T	-3.2 [†] [-7.8, +1.2]	-1.2 [†] [-5.5, +3.0]	-3.2 [†] [-7.2, +0.5]	+5.0 [†] [+0.2, +10.0]	-0.8 [†] [-5.0, +3.8]	-1.5 [†] [-5.5, +2.5]
	4P	+27.2 [+22.8, +32.0]	-1.2 [-3.0, +0.2]	-1.5 [-3.0, -0.2]	+3.8 [+1.2, +6.2]	-1.2 [-2.8, +0.0]	+1.2 [-0.8, +3.0]
	4T	+8.5 [†] [+3.2, +14.2]	+24.8 [†] [+19.5, +30.2]	+10.2 [†] [+5.5, +15.0]	+1.8 [†] [-3.5, +7.2]	+24.8 [†] [+19.8, +30.0]	+12.2 [†] [+7.2, +17.0]
Edge exists	1.7P	+4.0 [-0.2, +8.2]	+3.8 [-0.5, +8.0]	+5.0 [+0.2, +9.5]	+11.5 [+6.2, +16.5]	-8.2 [-11.8, -4.8]	-15.8 [-20.0, -11.8]
	1.7T	-2.5 [-4.5, -0.8]	-1.0 [-2.8, +0.5]	-1.2 [-3.0, +0.5]	-1.8 [-3.8, +0.0]	-0.8 [-2.5, +0.8]	-0.5 [-2.0, +1.0]
	4P	-4.8 [-7.8, -1.5]	+4.0 [+1.0, +7.0]	-10.0 [-13.8, -6.2]	-10.0 [-14.0, -6.0]	-2.8 [-5.8, +0.2]	-7.0 [-10.5, -3.5]
	4T	+0.0 [-0.8, +0.8]	-0.8 [-1.8, +0.0]	-1.2 [-2.5, +0.0]	-1.0 [-2.2, +0.0]	-0.5 [-1.5, +0.5]	+0.0 [+0.0, +0.0]
Node count	1.7P	+6.8 [+3.2, +10.5]	+67.2 [+64.5, +70.0]	+2.0 [-3.2, +6.8]	+47.0 [+42.5, +51.2]	+21.2 [+16.2, +26.2]	+39.8 [+36.0, +43.5]
	1.7T	+5.5 [+2.5, +8.5]	+9.0 [+6.5, +11.8]	+9.0 [+6.5, +11.8]	+9.0 [+6.5, +11.8]	+1.5 [-2.2, +5.0]	+9.0 [+6.5, +11.8]
	4P	+8.0 [+5.2, +10.8]	+8.8 [+6.0, +11.5]	+9.2 [+6.5, +12.0]	+9.8 [+7.2, +12.5]	+9.0 [+6.5, +11.8]	+9.2 [+6.8, +11.8]
	4T	+0.2 [-0.5, +1.2]	+0.2 [-0.5, +1.2]	+0.2 [-0.5, +1.2]	+0.5 [+0.0, +1.2]	+0.0 [-1.0, +1.0]	+0.5 [+0.0, +1.2]
Cycle	1.7P	-17.5 [†] [-23.0, -12.2]	+8.5 [+5.8, +11.2]	-0.2 [-4.2, +3.5]	+7.0 [+4.0, +10.2]	+2.0 [-1.5, +5.3]	+3.0 [-0.8, +6.8]
	1.7T	-21.8 [†] [-26.5, -17.5]	-0.5 [-2.8, +1.8]	+2.2 [+0.5, +4.0]	-0.8 [-3.2, +1.8]	-7.5 [-11.0, -4.2]	+3.0 [+1.5, +4.8]
	4P	-2.2 [-4.2, -0.5]	+0.5 [-0.5, +1.8]	-0.2 [-1.8, +1.2]	-0.2 [-1.8, +1.0]	-2.0 [-4.0, -0.2]	-0.5 [-2.0, +0.8]
	4T	-5.5 [-8.0, -3.0]	+0.0 [-1.2, +1.2]	+0.2 [-0.8, +1.5]	-0.5 [-2.0, +0.8]	-3.2 [-5.2, -1.2]	+0.5 [-0.5, +1.5]

Table 5: Every main-sweep effect against no primer (points) with its 95% bootstrap interval over graphs, 400 paired graphs per cell ($p \leq .50$). [†]: at least 15% truncation on either side. Node count (always 40) and cycle check (always yes) cannot measure reading the graph at 40 nodes. P/T: plain/thinking.

helps it (at most 2 points). At $p = .85$ it still answers 39, the degree of a node joined to all others, on 70.8% of finished responses. Plain Qwen3-4B’s wrong look-ups copy a nearby line: 17 of 57 equal the degree printed one sentence away, against 10.4 expected by chance ($p = .017$). Adding clustering and RWSE to the same degrees costs dense plain Qwen3-4B the most (Figure 3): pooled over $p \geq .65$, degree alone gains 9.3 points ($q = .03$) and all three statistics lose 21.3 ($q < .001$).

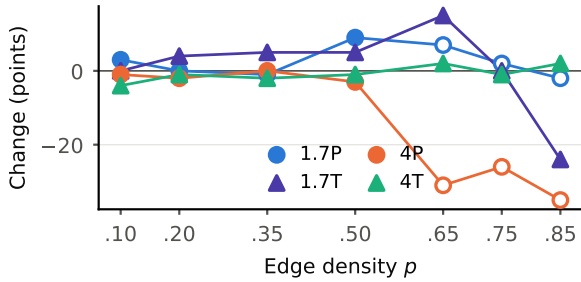


Figure 3: What adding clustering and RWSE to the same degrees changes on node degree (all three statistics minus the degree primer, points), per arm and density. Hollow as in Figure 2.

Filler and fragile questions (§4.3). A primer changes 38.5-69.1% of the questions filler changes and 3.1-15.2% of the rest. Some questions are fragile: on plain Qwen3-1.7B’s node degree, 24 of 400 change outcome when the same prompt is generated again. Every condition changes them far more often than the rest (45.8-66.7% against 14.4-26.1%, $q \leq .018$). Yet they carry only 11 to 16 of each condition’s 65 to 113 changes, a lower

bound on chance with only two generations per prompt. Where accuracy without a primer is 25-75%, components, clustering and RWSE average +1.6 points while filler averages -6.9 on the same cells.

Side statistics (§4.3). On edge existence, every primer that moves balanced accuracy does so in a plain arm (bold in Table 6). Plain Qwen3-4B’s false alarms also climb with shared neighbors, from 0.2% of non-edges with none to 38.8% with four or more. At high density the statistics barely vary within a graph: printed clustering values have SD 0.021 at $p = .65$, and rounded RWSE takes 3.30 distinct value pairs per graph at $p = .50$, against 12.50 distinct degrees. Gains there, such as clustering’s +11.3 on dense plain Qwen3-4B’s node degree with its procedure unchanged, are presentation effects.

Against the thinking mode (§4.4). Equivalence in Table 2 rests on 90% intervals. The component sentence’s gap on plain Qwen3-4B’s neighbor sets lies in $[-2.0, +2.5]$, inside the ± 3 -point margin. The degree primer’s gap on plain Qwen3-1.7B’s neighbor sets is not significant ($q = .24$), but its interval reaches +6.2, so the two are not shown to be level.

Thinking mode (§4.5). Thinking models check stated values against their own count. Under the degree primer, thinking Qwen3-1.7B still lists the neighbors in 90% of responses at $p \geq .35$, and thinking Qwen3-4B ends on the stated value in

Arm	Primer	Raw	Balanced	Yes
1.7P	none	70.50	79.86	56.2
	degree	74.50	82.30	51.7
	clustering	74.25	82.42	52.5
	RWSE	75.50	82.68	50.1
	all	82.00	87.71	44.8
	components	62.25	74.23	64.5
1.7T	filler	54.75	69.11	72.0
	none	99.00	99.02	27.1
	degree	96.50	97.61	29.9
	clustering	98.00	98.63	28.6
	RWSE	97.75	98.17	28.3
	all	97.25	98.12	29.5
4P	components	98.25	98.81	28.3
	filler	98.50	98.68	27.8
	none	90.50	93.22	35.7
	degree	85.75	90.27	40.7
	clustering	94.50	96.25	32.3
	RWSE	80.50	86.69	46.3
4T	all	80.50	86.69	46.1
	components	87.75	91.64	39.0
	filler	83.50	88.74	43.2
	none	99.75	99.83	26.8
	degree	99.75	99.83	26.8
	clustering	99.00	99.32	27.0
4P	RWSE	98.50	98.68	26.9
	all	98.75	99.15	27.1
	components	99.25	99.49	27.0
	filler	99.75	99.83	26.8

Table 6: Edge existence, main sweep (400 prompts per row): raw and balanced accuracy over all prompts and the share of finished answers that say yes, all in %. Truncation is in Table 8. Bold: balanced accuracy differs from the same arm without a primer (paired permutation, BH $q < .05$ within arm).

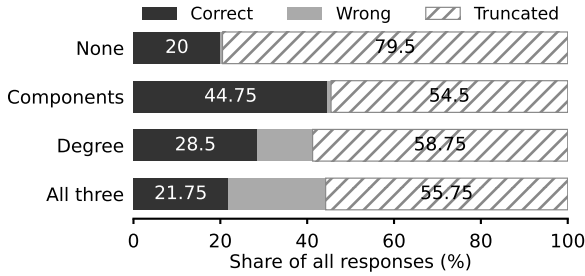


Figure 4: Thinking Qwen3-4B on edge count: correct, wrong and truncated shares of *all* responses. Conditions from top: none, components, degree, all three.

93% of its conflicts. In the follow-ups, clustering gains thinking Qwen3-1.7B 2.5 points at $p \leq .50$ ($q = .043$) and loses 5.4 at $p \geq .65$ ($q = .0015$), where filler loses 11.4. Of the answers a condition changes, 44.4-64.5% for thinking Qwen3-1.7B and 83.3-95.2% for thinking Qwen3-4B involve a response that ran out of tokens; Figure 4 shows this for thinking Qwen3-4B’s edge count.

What responses rest on (§4.6). Clustering and RWSE change the share of queries answered right

Arm	Primer	Correct “yes” (n)	Cycle named			No cycle	
			Real	Invented	Not a cycle	All	Edge count
1.7P	none	363	36.9	43.5	16.0	3.6	0.8
	degree	293	13.0	20.5	1.4	65.2	55.6
	clustering	397	39.5	48.1	7.8	4.5	1.0
	RWSE	362	33.7	40.3	19.9	6.1	1.7
	all	391	38.4	50.4	7.9	3.3	1.3
	components	371	27.8	40.4	25.3	6.5	4.3
1.7T	filler	375	33.3	55.5	9.6	1.6	0.0
	none	388	79.9	17.8	1.0	1.3	0.5
	degree	301	61.5	29.6	0.0	9.0	9.0
	clustering	386	80.6	19.2	0.0	0.3	0.3
	RWSE	397	68.0	20.7	10.6	0.8	0.5
	all	385	73.0	21.8	3.1	2.1	1.8
4P	components	358	74.9	22.6	0.0	2.5	2.0
	filler	400	77.2	19.8	2.5	0.5	0.0
	none	396	34.1	23.0	6.8	36.1	25.0
	degree	387	13.4	3.9	2.1	80.6	65.6
	clustering	398	48.7	31.2	1.0	19.1	5.8
	RWSE	395	33.2	20.8	38.7	7.3	0.5
4T	all	395	45.1	33.4	6.6	14.9	0.8
	components	388	39.2	18.6	3.4	38.9	26.5
	filler	394	44.2	28.2	5.3	22.3	2.3
	none	397	96.5	3.3	0.0	0.3	0.0
	degree	375	90.9	5.1	0.0	4.0	2.7
	clustering	397	95.5	4.5	0.0	0.0	0.0
4P	RWSE	398	55.0	2.5	42.5	0.0	0.0
	all	395	92.7	6.3	0.0	1.0	1.0
	components	384	92.2	4.4	0.0	3.4	2.9
	filler	399	94.2	5.0	0.5	0.3	0.0

Table 7: What the correct cycle-check answers rest on (main sweep; every correct answer is “yes”). *Correct (n)*: count of correct, finished answers; the other columns are % of those n , and Real, Invented, Not a cycle and All sum to 100. *Real*: every step of the cycle is an edge; *Invented*: a step is not an edge; *Not a cycle*: a length-2 or retraced walk. *Edge count*: the answers under All that argue from the number of edges, identified by their wording. The valid arguments are Real and Edge count, which is valid because 40 nodes with at least 40 edges must contain a cycle. Bold (Real, Invented only): differs from the same model without a primer (paired, BH $q < .05$).

on both degree and neighbor set with $q \geq .16$ in every arm. The degree primer lowers it for plain Qwen3-4B and raises it for plain Qwen3-1.7B, from 54.00% to 62.50%. Table 7 classifies every correct, finished cycle-check answer by the argument it cites. The degree primer raises plain Qwen3-4B’s valid share from 58.5% to 76.5%. Truncation accounts for 21.2 of the 21.8 points it costs thinking Qwen3-1.7B but only 8.2 of the 17.5 it costs plain Qwen3-1.7B (Tables 5 and 8). RWSE, all three statistics and filler lower plain Qwen3-4B’s valid share to 33.2%, 45.2% and 45.8% ($q < .05$).

Task	Arm	none	degree	clustering	RWSE	all	components	filler
Degree	1.7P	0.00	+0.0	+0.0	+0.2	+0.0	+0.0	+0.0
	1.7T	0.75	+4.5	-0.5	+0.0	-0.5	-0.2	-0.5
	4P	0.00	+0.0	+0.0	+0.0	+0.0	+0.0	+0.0
	4T	2.75	-2.5	-2.0	-0.5	-1.0	-0.5	-1.8
Neighbors	1.7P	0.00	+0.0	+0.0	+0.0	+0.2	+0.0	+0.0
	1.7T	1.25	+0.2	-1.0	-0.8	-0.5	-0.2	-0.8
	4P	0.00	+0.0	+0.0	+0.0	+0.0	+0.0	+0.0
	4T	3.25	-0.2	+1.2	-0.2	+1.0	+0.2	-0.5
Edge count	1.7P	29.75	-20.2	-11.8	-11.8	-1.8	-2.5	-3.5
	1.7T	81.75	-0.8	+0.8	+5.0	-12.2	+0.5	+1.2
	4P	0.00	+1.8	+0.2	+0.5	+0.8	+0.8	+0.0
	4T	79.50	-20.8	-26.0	-12.2	-23.8	-25.0	-13.0
Edge exists	1.7P	0.00	+0.0	+0.0	+0.2	+0.0	+0.0	+0.0
	1.7T	0.25	+0.2	+0.0	+0.0	-0.2	+0.0	-0.2
	4P	0.00	+0.5	+0.0	+0.0	+0.2	+0.0	+0.0
	4T	0.25	+0.0	+0.8	+1.2	+1.0	+0.5	+0.0
Node count	1.7P	0.00	+0.0	+0.0	+0.0	+0.0	+0.0	+0.0
	1.7T	7.75	-5.0	-7.8	-7.8	-7.8	-0.5	-7.8
	4P	0.00	+0.0	+0.0	+0.0	+0.0	+0.0	+0.0
	4T	0.00	+0.2	+0.2	+0.2	+0.0	+0.0	+0.0
Cycle	1.7P	9.00	+8.2	-8.2	-7.2	-8.0	-2.5	-2.8
	1.7T	3.00	+21.2	+0.5	-2.2	+0.8	+7.5	-3.0
	4P	0.75	+0.8	-0.2	+0.5	-0.2	+0.8	+0.5
	4T	0.75	+4.8	+0.0	-0.2	+0.5	+3.2	-0.5

Table 8: Complete main-sweep truncated shares: no-primer baseline (percent of all prompts) and paired changes (percentage points), for the same queries as Table 5. Each of its cells’ *wrong* share equals 100% minus its correct and truncated shares. P/T: plain/thinking; components and filler are diagnostics. No significance marks are assigned to truncation contrasts.

Node count. The encoding’s first line lists the 40 node ids, so node count measures which number a model writes, not reading. Plain Qwen3-1.7B answers 39, the last id, on 68.5% of finished responses without a primer and on 1.2% under clustering, hence Table 5’s large node-count effects.

C Pilot and the Clustering Screen

Pilot. On the published GraphQA graphs (5-19 nodes), Gemma 4 E4B/12B and Qwen3-8B/14B, each in plain and thinking mode, were near ceiling without a primer on most tasks; only edge count separated them. This is why the main study uses 40-node graphs and smaller models.

The clustering effect. Clustering on plain Qwen3-1.7B’s node degree emerged from a 40-cell screen, so Table 9 tests it on graphs the screen did not see, on a fresh seed, against filler, on 20-160-node graphs at fixed mean degree, at high density and on Qwen3-8B. It shows little sign of using the stated values. No response mentions clustering, and the gain does not depend on the queried node’s printed coefficient ($p = .5$). It falls on queried nodes early in the node list: +5.01, +0.75 and -0.11 points over the first, middle and last third at all seven densities, and 1.74 points less per 10 positions after adjusting for degree, clustering and density ($p = .018$); filler and components show no such gradient. It fixes 224 copying errors in the

Test	Gain	p or q
Screened graphs (700)	+1.29	$p = .47$
Unscreened graphs (2,100)	+2.1	$p = .031$
$p \leq .50$ (1,600 pairs)	+3.8	$q = .0035$
Fresh seed	+4.2	$p = .00055$ (.022 corr.)
Against filler	+3.2	$q = .014$
Fixed mean degree (3,200)	+6.2	$q < .001$
$p \geq .65$ (1,200 pairs)	-0.7	$q = .57$
Qwen3-8B (3,200)	-1.0	$q = .05$

Table 9: The clustering effect on node degree against no primer (against filler where the row says so), in points, for plain Qwen3-1.7B except the last row, which is Qwen3-8B. *Screened*: graphs the screen also used; *fixed mean degree*: the 20-160-node grid. *corr.*: Bonferroni over the screen’s cells.

neighbor list and introduces 176. Components and filler also cut the neighbors the model misses (0.94 and 1.05 per response at $p = .35$ -.50, against 0.86 with clustering and 1.49 with none).